HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Department of Actuarial Mathematics and Statistics

F78AA

ACTUARIAL AND FINANCIAL MATHEMATICS A

Semester 1 2016/17

Duration: Two Hours

Total marks: 60

ANSWER ALL QUESTIONS

Electronic calculators approved by the University may be used.

1. You are given the following information in respect of a pension fund:

Calendar year	Value of fund at 1 January	Value of fund at 30 June	Net cash flow received on 1 July
2013	£560 000	£685 000	£54 000
2014	£612 000	£650 000	£41 000
2015	£798 000	£864 000	£89 000
2016	£986 000		

Calculate the annual effective time-weighted rate of return earned on the fund over the period from 1 January 2013 to 1 January 2016.

[Total 4 Marks]

2. An investor purchases a bond on the issue date at a price of £96 per £100 nominal. Coupons at an annual rate of 4% are paid annually in arrears. The bond will be redeemed at par twenty years after the issue date.

Calculate the gross redemption yield from the bond.

[Total 4 Marks]

3. The rent on a property is of amount £r_1 and is payable in exactly one year's time. Rents are expected to grow by g per annum compound every year thereafter. Rent is payable annually on the property, in perpetuity.

Let £V_0 be the present value of the property and let i be the property owner's required annual effective return after income tax has been paid on the rent. The owner pays income tax at the constant rate of d, where $d \in (0, 1)$.

Writing down all steps, show that

$$V_0 = \frac{r_1(1-d)}{i-g}.$$

[Total 5 Marks]

4. (a) Writing down all steps, show from first principles that

$$(Ia)_{\overline{n}|} = \frac{\ddot{a}_{\overline{n}|} - nv^n}{i}.$$

[4 marks]

(b) Calculate the present value, at a rate of interest of 3% per annum effective, of an annuity where 10 is paid at the end of the first year, 12 is paid at the end of the second year, 14 is paid at the end of the third year and so on, with payments increasing by 2 per annum until the payment stream ends at the end of 20 years.

[6 marks]

[Total 10 Marks]

5. A loan of £100 000 is repayable over 25 years by level monthly instalments in arrears of capital and interest. The repayments are calculated using an effective rate of interest of 4% per annum.

(a) Show that the monthly repayment is £523.91. Minor discrepancies due to rounding errors will not be penalised.

[2 marks]

(b) Calculate the capital repaid in the first monthly instalment.

[2 marks]

(c) Calculate the total amount of interest paid during the last five years of the loan.

[4 marks]

(d) Calculate the interest included in the final monthly payment.

[2 marks]

(e) Without doing any further calculations, explain in words how the total annual repayment would alter if, under the original terms of the loan, repayments had been made less frequently than monthly.

[2 marks]

[Total 12 Marks]

6. Anna decides to boost her income by selling renewable energy that will be generated by her house. At the start of her energy project, Anna will pay a single lump sum of amount £8 000 to buy and install solar panels on her house. Sourcing and installing the solar panels will take one year.

The solar panels will then start operation and Anna will be able to sell her surplus energy to a local electricity company. Anna will receive payments from the sale of the surplus energy at the end of each year that the panels are in operation. Payments are received for 20 years. The amount of payment at the end of the first year of operation of the panels will be £300, increasing by 4% per annum compound thereafter. Additionally, Anna will receive a government grant of £150 for each year that the solar panels are in operation, paid at the start of each year that the panels are in operation. The solar panels will be redundant after 20 years of operation, after which no income will be received from either them or the government grant.

(a) Calculate the net present value of Anna's renewable energy project, at the start of the project. Assume an interest rate of 3% per annum effective.

[10 marks]

(b) Explain whether the internal rate of return achieved on this project will be greater or less than 3% per annum effective.

[2 marks]

[Total 12 Marks]

7. An investor borrows £120 000 at an effective interest rate of 6% per annum. The investor uses the money to purchase an annuity of £10 000 per annum payable half-yearly in arrears for 30 years. The annuity payments are applied to pay off the loan as soon as possible.

Once the loan is paid off, the investor can earn interest at an effective rate of 4% per annum on money invested from the annuity payments.

(a) Determine the discounted payback period for this investment.

[5 marks]

(b) Determine the profit the investor will have made at the end of the term of the annuity.

[8 marks]

[Total 13 Marks]

END OF PAPER